ETH zürich

Safety, Environmental Assessment and Risk Analysis

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Lecture 3

Explosions

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Outline and learning outcomes

1. Introduction to explosions [1,2]
2. Mechanical explosions [1]
3. Confined vapor and dust cloud explosions [1]
4. Boiling liquid expanding vapor explosion (BLEVE) [1]
5. Unconfined vapor cloud explosion (UVCE) [1]
6. Damage of explosions [1,2]

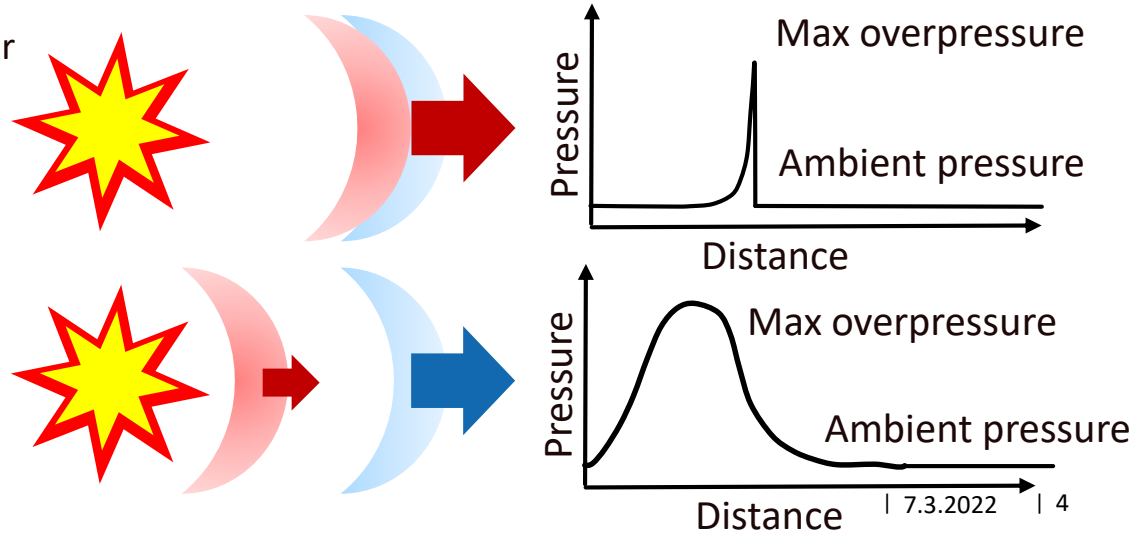
Learning outcomes

- Gain an understanding of the fundamentals of explosions in chemical systems
- Gain an understanding of the main types of explosions and their potential damage
- Calculate the damage of explosions

Introduction to explosions (I)

- Explosions are the most destructive and damaging events that can take place in chemical processes.
- An explosion is a rapid expansion of gases resulting in a rapidly moving pressure or shock wave.
  - The expansion can be mechanical (by means of a sudden rupture of a pressurized vessel), or it can be the result of a rapid chemical reaction.
  - Combustion reactions (the most common cause of explosions) only take place if the fuel concentration is between the Lower and Upper Flammability Limits.
- The chemical explosion encompasses a reaction front (where fuel is still reacting, generating heat and flue gases) and a shock/pressure front (volume of air and gas displaced by the explosion). The relative speed of these fronts differentiate explosions between deflagrations and detonations:

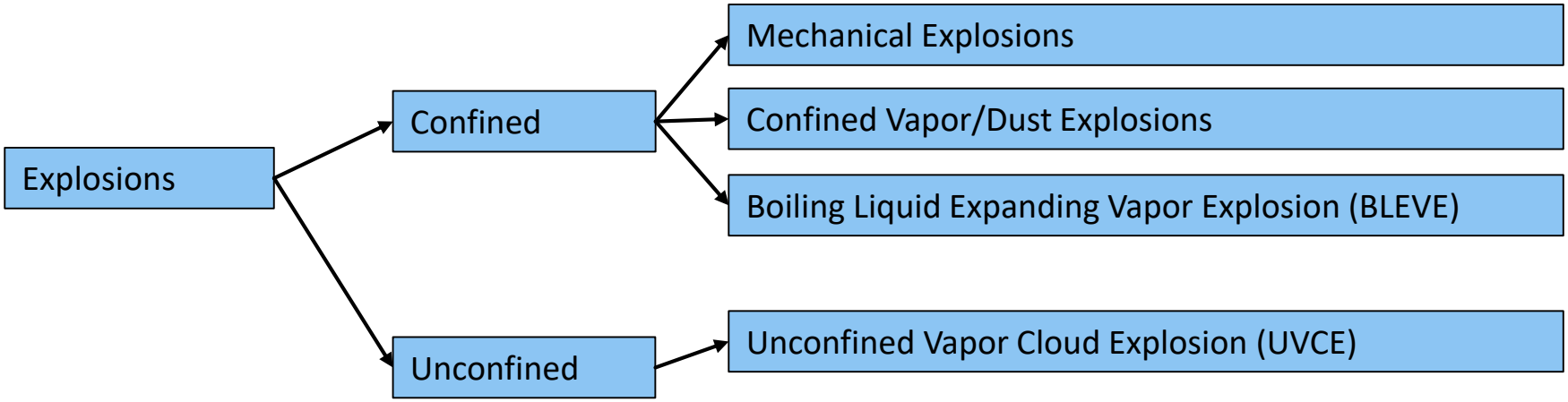
- Detonation: The reaction front moves at or beyond the speed of sound (~331 m/s), pushing the pressure front.



- Deflagration: The pressure front moves at the speed of sound, while the reaction front moves slower.

Introduction to explosions (II)

- The behavior of explosions heavily depends on environmental and physical factors (for example, ambient temperature pressure and humidity, the geometry of the surroundings, distribution of fuel in explosive environment…).
- Explosions can be classified as confined or unconfined depending on where the explosion originates:

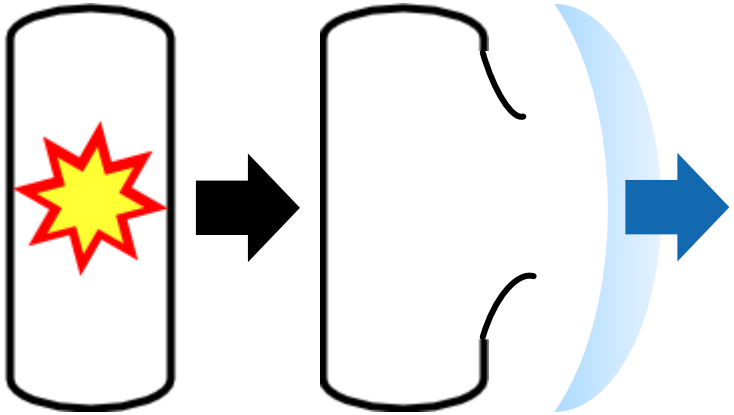


- Ultimately, explosions cause damage through three mechanisms:
  - Overpressure generated from the pressure blast
  - Shooting of projectiles
  - Thermal radiation (if the explosion generates heat, for example, through combustion)

Mechanical explosions

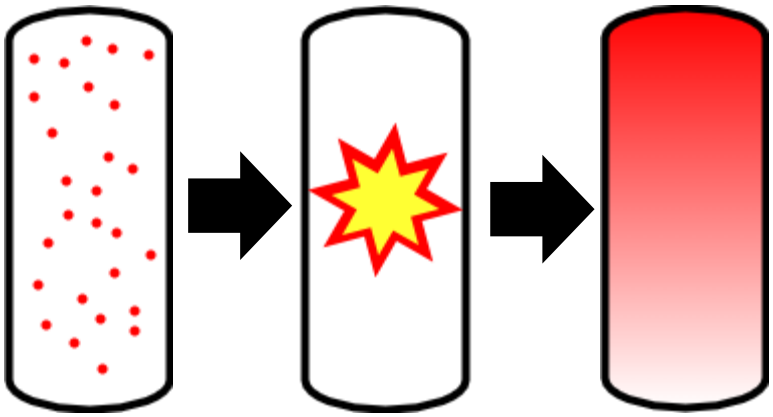
- A reaction does not occur and the energy is obtained from the substance directly, e.g., a compressed gas that is suddenly released, excessive vessel loading, etc.

- The maximum pressure reached exceeds the critical design pressure of the containing vessel, so the vessel will rupture, releasing high-pressure material (potentially flammable).



Confined vapour and dust cloud explosions

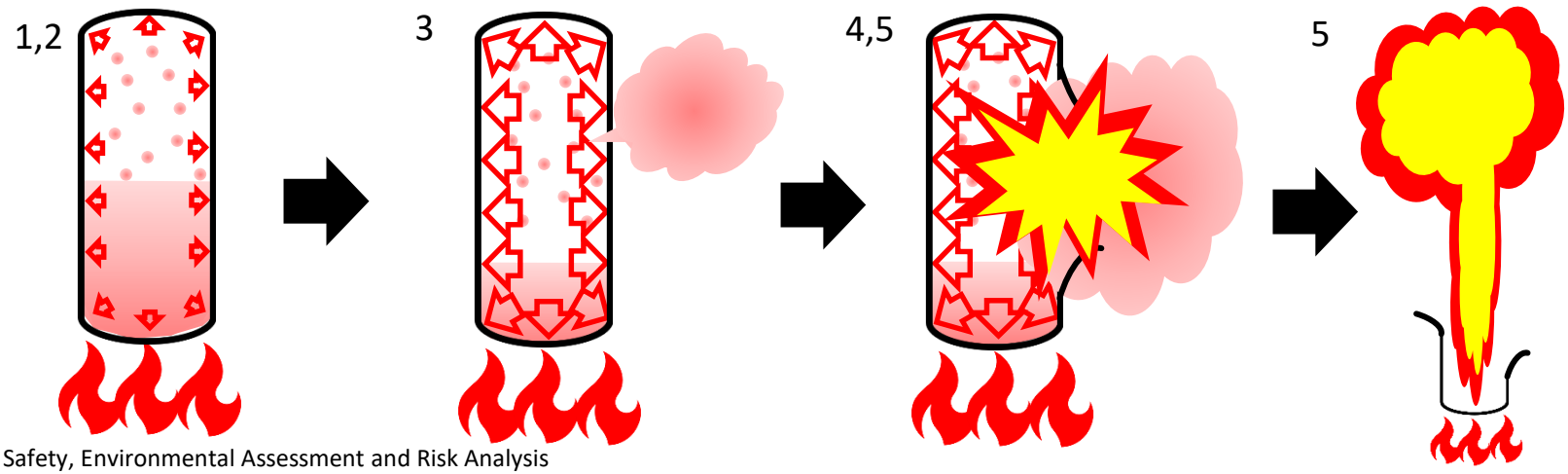
- Confined gas, vapor and dust cloud explosions take place inside storage vessels, tanks or other confined spaces (e.g., flour mills, buildings).



- Explosions only take place if combustion occurs, that is, if the fuel concentration is between the lower and upper flammability limits (LFL and UFL).
- Confined explosions often assume that the containment does not break in the explosion (constant volume).

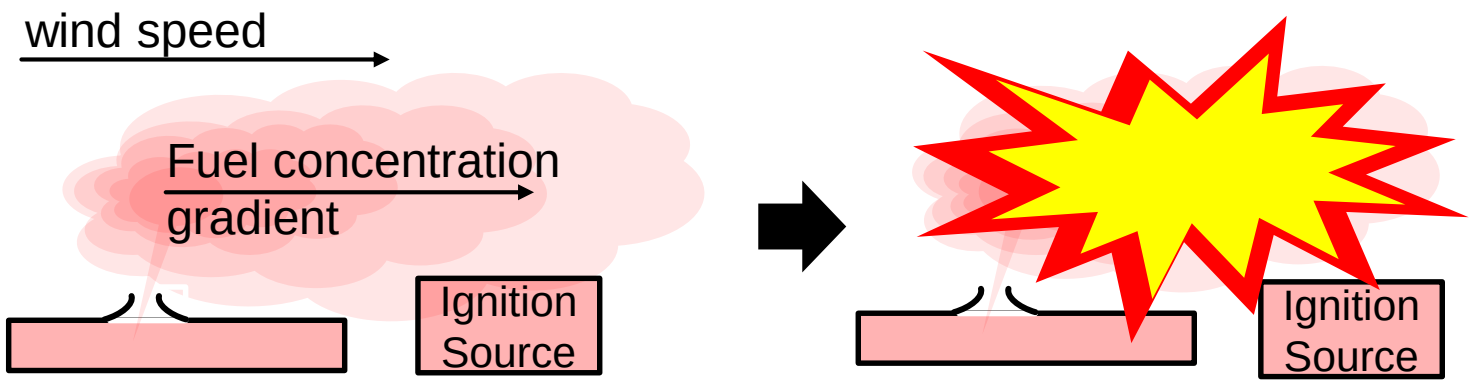
Boiling Liquid Expanding Vapor Explosion (BLEVE)

- BLEVEs are one of the most common types of explosions. A BLEVE is the explosive vaporization of a large fraction of the vessel content, often followed by combustion or explosion of the vaporized cloud if it is flammable. It happens when a tank containing a liquid (often pressurized) at a temperature above its atmospheric boiling point ruptures.
- A BLEVE takes place in 5 steps:
  1. An external fire or heat source develops adjacent to a tank containing a liquid.
  2. The fire heats up the walls of the tank.
  3. The liquid vaporizes and increases the pressure inside the vessel.
  4. The generated flammable vapor builds up in the vessel, increasing its temperature until it finds a way out (e.g., a safety valve or a leak resulting from the vessel losing its structural strength).
  5. The tank ruptures, explosively vaporizing its content, which may ignite as the tank ruptures or lead to a vapor cloud explosion when it finds an ignition source in the surroundings.



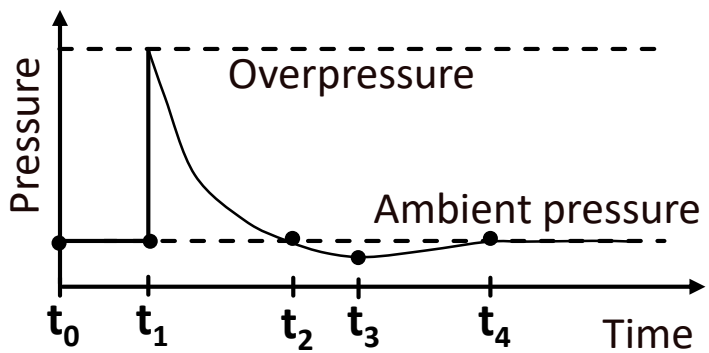
Unconfined Vapor Cloud Explosion (UVCE)

- Unconfined Vapor Cloud Explosions (UVCE) are the result of a flammable gas release. They represent the most dangerous and destructive explosions. It comprises three steps:
  1. A sudden release of flammable vapor (often due to the rupture of a tank containing a superheated and pressurized liquid)
  2. Dispersion of the vapor throughout the plant site while mixing with air
  3. Ignition of the resulting vapor cloud
- Unconfined explosions are rarer than confined explosions because the explosive material is frequently diluted below the LFL by wind dispersion.
- These explosions are destructive because large quantities of gas are frequently involved, affecting large areas.
- They should be prevented by keeping low inventories of volatile, flammable materials and applying process conditions that minimize flashing if a vessel or pipeline rupture.



Damage of explosions: Pressure blast damage (I)

- The sudden release of energy in explosions causes air and gas around it to move at supersonic speeds, causing a shock wave.
- Shock waves are abrupt pressure waves moving through a gas. A shock wave in the open air is followed by a strong wind. The combination of a shock wave and the wind is called a blast wave. The pressure increase in the shock wave (overpressure) is so rapid that the process is assumed to be adiabatic.
- The diagram below shows the pressure evolution of a blast wave in a fixed location:



- t_0: Explosion takes place.
- t_1: Pressure front arrives.
- t_2: The shock wave quickly decreases in pressure as it passes, leaving a strong wind.
- t_3: The shockwave reaches the maximum under-pressure (below atmospheric pressure).
- t_4: The reference location goes back to ambient pressure.

Damage of explosions: Pressure blast damage (II)

- The overpressure achieved in the blast is proportional to the energy released in the explosion (e.g., the kg of TNT equivalent, m_{TNT}) but decreases with the distance from the explosion location (X, in m). The explosion overpressure (P_o, in kPa) can be estimated from the following equations:

$$P_o = P_a \frac{1616\left[1+\left(\frac{X_o}{4.5}\right)^2\right]}{\sqrt{1+\left(\frac{X_o}{0.048}\right)^2}\sqrt{1+\left(\frac{X_o}{0.32}\right)^2}\sqrt{1+\left(\frac{X_o}{1.35}\right)^2}}, \qquad \text{with } X_o = X \cdot m_{TNT}{}^{-1/3}$$

, where P_a is the ambient pressure (in kPa).

- To estimate the damage of a pressure blast in a given location (distance), proceed as follows:
  1. Compute the energy of the explosion using thermodynamic approaches
  2. Convert the energy into an equivalent TNT amount
  3. Use the empirical correlation to estimate the overpressure
  4. Use available data to estimate the damage

Damage of explosions: Pressure blast damage (III)

Equivalent mass of TNT (m_{TNT}, in kg TNT equivalent) is generally used to express the energy released in explosions, assuming that an exploding fuel mass behaves like exploding TNT on an equivalent energy basis.

Fuel combustion explosions

- The blast damage can be calculated using the energy released, expressed as TNT equivalent mass m_{TNT} (in kg TNT equivalent):

$$m_{TNT} = \frac{\alpha \cdot \Delta H_c \cdot m_F}{4.52 \cdot 10^6}$$

, where ΔH_c is the lower heating value of the flammable gas (in J/kg fuel), m_F is the mass of fuel (in kg fuel), and α is the empirical explosion efficiency (assume 10% unless specified otherwise).

Mechanical explosions

- The energy released in mechanical explosions E_M (in J) can be estimated with the following expression:

$$E_M = \frac{V(P_2 - P_1)}{(\gamma - 1)}$$

- ,where V (in m^3) is the volume of the vessel, γ is the heat capacity ratio, P_2 is the explosion overpressure, and P_1 is the ambient pressures (both in Pa). The heat capacity ratio can be assumed to be γ=1.4.
- The TNT equivalent mass (in kg TNT equivalent) for mechanical explosions can be calculated as follows (an efficiency factor could also be applied if available):

$$m_{TNT} = {E_M}/{4.52 \cdot 10^6}$$

Adapted from: Crowl, D.A., Louvar, J.F., 2011. Chemical process safety: fundamentals with applications, 3rd ed. Prentice Hall.

Damage of explosions: Pressure blast damage (IV)

- The following table provides damage estimates depending on the explosion overpressure:

Overpressure (P_o, in kPa)	Damages
0.3	Loud noise (143dB), sonic boom
2.1	Safe distance (only minor damages to structures and people)
7.1	Glass breaking*
13.8	Partial collapse of walls and roofs on houses
33.7	Serious structural damage*
48.2	Complete destruction of houses, derailing and displacement of train wagons
68.9	Destruction of buildings, displacement of very heavy equipment
101.3	Eardrum rupture*
183.4	Deaths caused by lung hemorrhage*

Adapted from: Crowl, D.A., Louvar, J.F., 2011. Chemical process safety: fundamentals with applications, 3rd ed. Prentice Hall.
* Estimated from Probit correlations (>95% probability) in table 2-5 from Crowl, D.A., Louvar, J.F., 2011. Chemical process safety : fundamentals with applications, 3rd ed. Prentice Hall.

Damage of explosions: Projectiles

- In addition to the direct overpressure damages, the explosion blast can throw parts of broken equipment and other objects as projectiles at high speeds and large distances.

- The maximum distance at which projectiles can land (d_{pr}, in m) can be estimated from the equivalent mass of TNT (m_{TNT}, in kg) with the following expression:

$$d_{pr} = 290.69 \cdot {m_{TNT}}^{0.217}$$

Adapted from figure 6-26 in: Crowl, D.A., Louvar, J.F., 2011. Chemical process safety: fundamentals with applications, 3rd ed. Prentice Hall.

- **Example:** Estimate the maximum distance at which projectiles can land from a BLEVE explosion involving 500 kg of gasoline (ΔH_c= 46.4 MJ/kg). Assume a 10% explosion efficiency.
- $m_{TNT} = \frac{\alpha \cdot \Delta H_c \cdot m_F}{4.52 \cdot 10^6} = \frac{(0.1) \cdot (46.4 \cdot 10^6) \cdot (500)}{4.52 \cdot 10^6}$ = 513.27 kg TNT equivalent
- $d_{pr} = 290.69 \cdot {m_{TNT}}^{0.217} = 290.69 \cdot (513.27)^{0.217}$= 1126.1 m

Projectile fragments will be thrown away at 1126.1 m from the explosion location.

Recap

- Fundamentals of explosions and their main types.
- Fundamentals of each explosion type.
- Estimation of the damage caused by explosions.

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